ARPITA MALLIK

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Chittagong, Bangladesh

Summary

I am a recent Computer Science and Engineering graduate. My research interests include Large Language Models, Natural Language Processing, Multilingual and Low-resource NLP, Parameter-Efficient Fine-tuning, and Trustworthy AI, especially in terms of factuality of LLMs, their sycophantic tendencies, deception, and safety concerns. I am especially interested in studying and improving the reliability and robustness of generative AI systems.

Education

- **Chittagong University of Engineering & Technology** — *March 2022 – July 2026*
  BSc. in Computer Science & Engineering — Chittagong, Bangladesh
  - GPA: 3.63/4.00 (most recent 78.25 credits: 3.87/4.00)

Research Experience

- **Research Intern, ELITE Research Lab** — *2026 – Present*
  Adv. Md Kishor Morol - Research Intern, Meta
  - Building a counterfactual multimodal deception benchmark for vision-language models.
- **Shared Tasks and Workshop Publications** — *2025 – 2026*
  - Led and contributed to multiple competitive shared tasks across multimodal misogyny detection and Bengali polarization detection.
  - Mentored multiple research projects, contributing to workshop papers accepted at ACL Anthology venues such as DravidianLangTech and LREC.

Thesis

- **Undergraduate Thesis** — *2026 – Present*
  An Emotion-Aware Large Language Model-based QA System for Women's Healthcare Information in Bangla
  - Curated a Bengali and code-mixed (Banglish) women's health QA dataset (2,700 pairs) with binary emotion labels (neutral/concerned) across eight topics including PCOS, menstrual health, pregnancy, menopause, and postpartum depression.
  - Built an emotion-aware QA system on a QLoRA (4-bit) fine-tuned Gemma-2-9B model, injecting FiLM-based emotion conditioning at decoder layers via forward hooks, driven by a transformer-based emotion classifier.
  - Ran controlled ablations to attribute performance gains to specific interventions rather than architectural novelty.
  - Extending the work with a trustworthiness evaluation using 102 health-myth minimal pairs with neutral vs. concerned framing and ground-truth stance labels; measuring directional stance flips (correct→wrong vs. wrong→correct) across baseline, LoRA, and FiLM systems, with automated stance labeling validated against human annotations.

Currently in Progress

- **Counterfactual Multimodal Deception Benchmark** — *2026 – Present*
  ELITE Research Lab
  - Designing each scenario against a frozen 2×4 schema (Image_Type: Original/Visual_CF × Prompt_Type: Baseline/Incentive/Oversight/Correction), scored across six deception categories: Fabrication, Omission, Sycophancy, Bluffing, Sandbagging, Obfuscation.
  - Building out the dataset across Food Safety, Safety & Security, Wildlife & Nature, and Traffic & Transportation domains.
- **Survey on Trustworthy Explainable AI (XAI) in Oncology** — *2026 – Present*
  - Co-developing a survey on trustworthy XAI in oncology and clinical decision support, structured around four core dimensions: security (adversarial attacks/defenses on explanations), robustness (perturbation and faithfulness evaluation), fairness (disparities across demographic subgroups), and clinical validation (how unstable explanations affect clinician trust).
  - Proposing an original five-dimension evaluation framework: explainability, security, fairness, robustness, and clinical validation.

PUBLICATIONS
C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

[C.1] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages.*

[C.2] **CUET-823 at SemEval-2026 Task 9: LoRA-Based Instruction-Fine-Tuned LLMs vs. Transformer Models for Bengali Polarization Detection**. Accepted to the *20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

TEACHING EXPERIENCE

- **ASSRO, CUET** — *2025 – 2026*
  Machine Learning Instructor — Chittagong, Bangladesh
  - Mentored junior students in Machine Learning and Deep Learning through technical sessions and hands-on workshops.
  - Guided research and implementation for NLP, Computer Vision and Multimodal shared-task projects, contributing to workshop paper publications at ACL Anthology venues.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **ML/DL Frameworks:** PyTorch, scikit-learn
- **NLP & LLMs:** Transformers, Hugging Face, QLoRA, LoRA, RAG, FiLM Conditioning
- **Deployment & Tools:** FastAPI, Flask, Docker, AWS EC2/S3, Git, GitHub Actions, MLflow, DVC

AWARDS & HONORS

Finalist – DL Sprint 4.0 (Top 10% / 192 teams, nationwide) — *February 2026*
BUET CSE Fest 2026

- Built a pipeline for Bangla long-form speech recognition and speaker diarization using Whisper-based ASR and fine-tuned diarization models.

Champion – NodeScape Hackathon (1^{st} place) — *July 2025*
CUET Computer Club

- Developed an AI-powered graph classification system with algorithm visualization; Dockerized CI/CD deployment using React, Flask, and Python.

Finalist – Intra CUET Machine Learning Contest — *July 2025*
IEEE CS CUET Student Branch Chapter

- Built a Bangla physics MCQ question-answering system leveraging LLMs for concept-based queries.

Top 7 Finalist – HackCSB 2024 (out of 116 teams) — *August 2024*
Coding Shikhbe Bangladesh

- Built "MotivationMate," a mental health-focused web application.

LEADERSHIP & SERVICE

Vice Chair (Machine Learning Wing) — *July 2025 – Present*
IEEE CS CUET Student Branch Chapter